BCSC0008: COMPUTER NETWORKS

Objective: *The objective is to understand fundamental underlying principles of computer networking, details and functionality of layered network architecture.*

Credits:04 **L-T-P-J:3-1-0-0**

Module No.	Content	Teaching Hours
I	**Introduction Concepts:** Goals and Applications of Networks, Network structure and architecture, The OSI reference model, services, Network Topology Design, Physical Layer Transmission Media, Line coding scheme, switching methods (circuit switching, Packet switching), TDM. **Medium Access sub layer:** Medium Access sub layer - Channel Allocations, LAN protocols - ALOHA protocols, CSMA, CSMA/CD, Overview of IEEE standards	13
II	**Data Link Layer –**Error detection and correction, Flow control (sliding window protocol) **Network Layer:** Network Layer –IP addressing, subnet, CIDR, VLSM, Internetworking, Address mapping, routing. Connecting devices.	14
III	**Transport Layer:** Transport Layer - Design issues, connection management, Flow control, TCP window management, congestion control-slow start algorithm **Application Layer:** Data compression, Data Encryption, File Transfer, DNS, HTTP, SMTP, TELNET. Introduction to IPv6, transition from IPv4 to IPv6.	13

Text Books:

- Forouzan B. A. (2004), "*Data Communication and Networking"*, 4th Edition, McGrawHill.

References:

- Kurose, J.F. and Ross K.W. *(*2005), *"Computer Networking: A Top-Down Approach Featuring the Internet"*, 3rd Edition, Addison-Wesley.
- A.S.Tanenbaum (2006), "*Computer Networks"*, 2nd Edition, Prentice Hall India.

Outcome: After the completion of the course, the student will be able to:

- Understand the terminology and concepts of the OSI reference model and the TCP/IP reference model.
- Apply the concepts of protocols, network interfaces, and design/performance issues in local area networks and wide area networks.